Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 14

Starting Your Cloud Journey

Choosing a cloud platform, the challenges, cloud native and where to start.



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Recap

LAST SESSION

DevOps: CI/CD

Continuous integration, delivery and deployment, and the pipeline that runs them.

LAST SESSION

IaC

Infrastructure defined as configurations, kept in source control.

Overview today

1. How to choose a cloud platform?
2. Challenges
3. Cloud journey

THIS WEEK

Your cloud journey

The last session: how to pick a platform, what to watch for, and where to go next.

PART 1

How to choose a cloud platform?

Some parameters to consider while choosing a cloud platform.

Week 14 · 04

Criteria

Seven parameters to consider:

1. Company culture
2. Your background
3. Project needs
4. Trust
5. Service level agreements
6. Compliance
7. Partner ecosystem



Criteria: culture and background

01

Company culture

More often than not, choosing a cloud platform is not a technical decision but a cultural one. It depends on the company's culture, the tools and systems it uses, and the programming languages and frameworks developers are familiar with.

02

Your background

You might be unable to influence which platform a company picks. But you have preferences in the languages and systems you know or want to work with. Pick a platform based on these, then find companies that work with it.

Criteria: project needs and trust

03

Project needs

“Pick the right tool for the job.” Every project is different and has different needs: technical, legal, financial or anything else. The choice of cloud platform should reflect the needs of the particular project.

04

Trust

Putting the company’s data and applications on the cloud is a big decision. Consider the vendor’s history, service and support, how secure the platform is, how they handle data privacy and security incidents, and how much they invest.

Criteria: SLAs and compliance

05

Service level agreements

SLAs are binding guarantees that cloud vendors provide for their services' uptime, a way of gauging reliability and availability. If a vendor cannot meet its SLA for a service, it is liable for compensating its customers.

06

Compliance

Businesses often have regulatory requirements to meet. Many are obliged to comply with GDPR, and the health industry with HIPAA. Check that the platform is compliant with the standards and regulations that matter to your business.

Criteria: partner ecosystem

07

Existing systems

Companies moving to the cloud do not discard their existing systems and start from scratch. It matters whether they will be able to continue to use the software and critical systems they already run.

HOW PLATFORMS HELP

The marketplace

All major cloud platforms have a thriving partner ecosystem. A marketplace makes it easy to discover, deploy and use applications, services and products from third-party vendors.

PART 2

Challenges

Complexity, scalability, availability, cost and local development.

Week 14 · 10

Challenges: complexity and scalability

Complexity

A lot of services are involved in building cloud solutions: both AWS and Azure offer well over 200. Building applications that are available, secure and can handle the load requires effort and understanding.

Scalability

For a database service: how much load can it handle, how many reads and writes, how do we grow it at the capacity limit, and how would we know it is under stress? Do we need different programming techniques to write scalable code?

Challenges: availability

The cloud offers the capability to build highly available solutions, but we need to *design* them that way.

A core tenet

When our solutions run in the cloud, we want to be sure our customers have access at all times.

Many services

Any cloud solution uses multiple services, internal and external, to build the final application.

The weakest link

If even one service in the chain goes down, the application will not function as expected.

Challenges: cost and local development

Cost

There is a perception that the cloud is cheaper because we only pay for what we consume. That might be true, but we cannot take it for granted. The cloud can get expensive if we are not mindful. For every solution we design, cost must be factored in.

Local development

Not all services can run on a local machine. Many PaaS services need connectivity to the cloud, which makes testing and debugging harder. Emulators cover some common services. Keep a separate cloud environment for development and testing.

PART 3

Cloud native

Software designed, developed and operated to take full advantage of the cloud computing model.

Week 14 · 14

Cloud native

Cloud-native applications are *built to be run in the cloud*.

We leverage the capabilities of the cloud to build resilient, highly scalable and highly available applications.

Technologies used: microservices, containers, container orchestrators, immutable infrastructure, APIs, Infrastructure as Code and source control management.

Cloud native in practice

Microservices in containers

Applications are often built as microservices running in Docker containers and orchestrated in Kubernetes.

CI/CD pipelines

Pipelines provide the necessary infrastructure, and push features and changes into production continuously.

How, not where

It is not tied to the cloud environment. We can use the cloud-native architecture in private clouds as well.

It is about how, and not where, applications are developed and deployed.

Architectural styles over time

Development process	Application architecture	Deployment and packaging	Application infrastructure
Waterfall	Monolith	Physical server	Datacenter
Agile	N-tier	Virtual servers	Hosted
DevOps	Microservices	Containers	Cloud

Each row is one era. The last row is cloud native.

Who can benefit from cloud native?

Cloud native is *not for every* organization or type of application.

It is a complicated and costly process, so we need a proper cost-benefit analysis to ensure it is a good fit. Proposing a cloud-native architecture in every situation could be overkill.

Where cloud native might be a good fit

Evolving features

Enterprise applications that need to constantly evolve their features over time.

High velocity

Applications that demand a high velocity of feature releases with greater confidence.

Small releases

Continuous, smaller, independent features released without affecting the entire application.

Many teams

Multiple teams on different technology stacks, each using the languages suitable for the job.

Independent scale

Applications where some components or features should scale independently.

No lock-in

Organizations that do not want to be locked in with a specific vendor.

Cloud-native tools and technologies

Open source

Kubernetes, Grafana, Istio, ELK Stack and Prometheus.

Cloud platforms

GCP, AWS and Azure all have managed Kubernetes services on their platforms.

Tools are not only provided by cloud platforms. Some of the most used are open-source projects with great adoption.

PART 4

Where to start?

The best way to learn about any technology is to practice it.

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Where to start?

01

Explore

Explore the products and services the major cloud platforms offer. If you have a clear picture of which platform to pick, start there.

02

Join a community

Find a group focused on cloud in your town or city and start attending. Technology is full of people eager to share their knowledge.

03

Practice

The best way to learn about any technology is to practice it.

Roadmap

roadmap.sh/devops

roadmap.sh/aws-best-practices

roadmap.sh/aws

Overview of AWS trainings and certifications (PDF, pages.awscloud.com)

AWS certification paths (PDF, d1.awsstatic.com)

Real job qualifications

Two current job postings, to see what employers ask for.

www.linkedin.com/jobs/view/4328655923

www.linkedin.com/jobs/view/4286645615

Source: LinkedIn · 24

Notable alumni

www.linkedin.com/in/yin-soknara

www.linkedin.com/in/kheang-sokuntheary-150634299/

Source: LinkedIn · 25

END OF THE COURSE

Good luck!

Pick a platform, join a community, and practice.